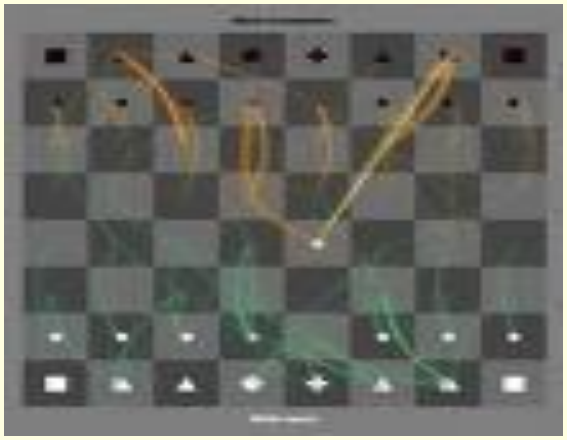




Artificial Intelligence (Part 2a)

Knowledge Representation and Search:

Propositional Logic

Course Contents

Again..Selected topics for our course. Covering all of AI is impossible!

Key topics include:

- Introduction to Artificial Intelligence (AI)
- Knowledge Representation and Search
- Introduction to AI Programming
- Problem Solving Using Search
- Exhaustive Search Algorithm
- Heuristic Search
- Techniques and Mechanisms of Search Algorithm
- Knowledge Representation Issues and Concepts
- Strong Method Problem Solving
- Reasoning in Uncertain Situations
- Soft Computing and Machine Learning

KR and search

- In the last lecture we considered the importance of 2 focus of AI research???
- Will look at logic as a method to formalize and reason about knowledge.
- Will consider the syntax and semantics of propositional logic.
- Will look at truth tables for compound propositions.

Ask google ..what is propositional logic?

- **propositional logic** logic
  (or "propositional calculus") A system of symbolic logic using symbols to stand for whole propositions and logical connectives. Propositional logic only considers whether a proposition is true or false. In contrast to predicate logic, it does not consider the internal structure of propositions.
  (2002-05-21)

Logic in General (1)

- *Logics* are formal languages for representing information such that conclusions can be drawn
- When people talk about logic they often mean *propositional* or *first-order predicate* logic.

Logic in General (2)

- A *logic* usually has a well defined syntax, semantics and proof theory.
- The *syntax* of a logic defines the syntactically acceptable objects of the logic, or *well-formed formulae*.
- The *semantics* of a logic associate each formula with a *meaning*.
- The *proof theory* is concerned with manipulating formulae according to certain rules.

Propositional Logic

- The syntax of propositional logic is constructed from *propositions* and *connectives*.
- A **proposition** is a statement that is either true or false but not *both*

Propositions???

- Najib is Prime Minister
- What is the time?
- 2 + 3 = 5
- ‘Phone’ has five letters.
- 2 + 3 = 6
- Oh dear!
- ‘Work’ has five letters.
- the reactor is on
- 2+3
- *All* elephants have 4 legs
- I like AI class

How to determine Propositions??

- It is possible to determine whether any given statement is a proposition by prefixing it with

  It is true that . . .

  and seeing whether the result makes grammatical sense.
- Propositions are often abbreviated using *propositional variables* eg p, q, r.
- Thus we must associate the propositional variable with its meaning i.e.

  Let p be *Najib is Prime Minister*.

Connectives

- Propositions may be combined with other propositions to form **compound propositions**. These in turn may be combined into further propositions.
- The connectives that may be used are

$\wedge$	and	conjunction (& or .)
$\vee$	or	disjunction (| or +)
$\neg$	not	negation (~)
$\Rightarrow$	if . . .then	implication ($\rightarrow$)
$\leftrightarrow$	if and only if	equivalence ($\Leftrightarrow$)

- Some books use different notations. Some of these are given in parentheses.

Propositional Logic: Syntax

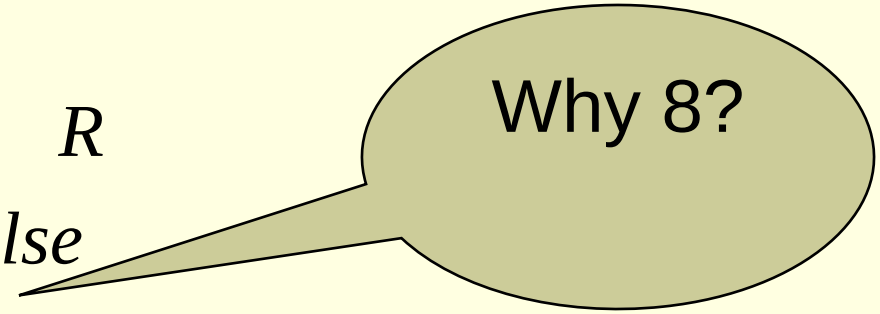
- The proposition symbols $P1$, $P2$ etc are sentences
- If S is a sentence, $\neg S$ is a sentence (*negation*)
- If S_1 and S_2 are sentences, $S_1 \wedge S_2$ is a sentence (*conjunction*)
- If S_1 and S_2 are sentences, $S_1 \vee S_2$ is a sentence (*disjunction*)
- If S_1 and S_2 are sentences, $S_1 \Rightarrow S_2$ is a sentence (*implication*)
- If S_1 and S_2 are sentences, $S_1 \Leftrightarrow S_2$ is a sentence (*equivalence*)

Propositional Logic: Semantic

Each model species true/false for each proposition symbol

E.g. P Q R

true *true* *false*



(With these symbols, 8 possible models, can be enumerated automatically.)

Rules for evaluating truth with respect to a model m are determined by *truth tables*

And $\wedge$ (also called *conjunction*)

- The *conjunction 'p AND q',* written $p \wedge q$, of two propositions is true when both p and q are true, false otherwise.
- We can summarise the operation of $\wedge$ using a truth table. Rows in the table give all possible setting of the propositions to true (T) or false (F).

p	q	

p	q	$p \wedge q$
T	T	T
T	F	F
F	T	F
F	F	F

Natural Language Meaning (1)

p	Its Monday.
q	Its raining.
$p \wedge q$	Its Monday and its raining.
	Its Monday but its raining.
	Its Monday. Its raining.

Natural Language Meaning (2)

Caution: semantics?

p	I took a shower
q	I woke up
$p \wedge q$	“I took a shower and I woke up”
$q \wedge p$	“I woke up and I took a shower”.

Logically the same! *WE* may see a difference.

The word *both* is often useful eg. *I both took a shower and I woke up*.

Or $\vee$

Also called *disjunction.*

The disjunction 'p OR q', written $p \vee q$, of two propositions is true when p or q (or both) are true, false otherwise.

Sometimes called *inclusive or*.

p	q	

p	q	$p \vee q$
T	T	T
T	F	T
F	T	T
F	F	F

Natural Language Meaning

p	It's Monday.
q	It's raining.
$p \vee q$	It's Monday or it is raining.

The word *either* is often useful eg. *either it's Monday or it is raining*.

It *also* includes the case of rain on a Monday!

Not ¬

Also known as *negation*

The negation 'NOT p' of a proposition (or $\neg p$) is true when p is false and is false otherwise. $\neg p$ may be read that it is false that p.

p	

p	$\neg p$
T	F
F	T

Negation is a *unary* connective. It only takes one argument.

Conjunction and disjunction were both *binary* connectives.

Natural Language Meaning

p	Logic is easy.
$\neg p$	It is false that logic is easy.
	It is not the case that logic is easy.
	Logic is not easy.

If . . . then $\Rightarrow$

Also known as **implication**

The implication 'p IMPLIES q', written $p \Rightarrow q$, of two propositions is true when either p is false or q is true, and false otherwise.

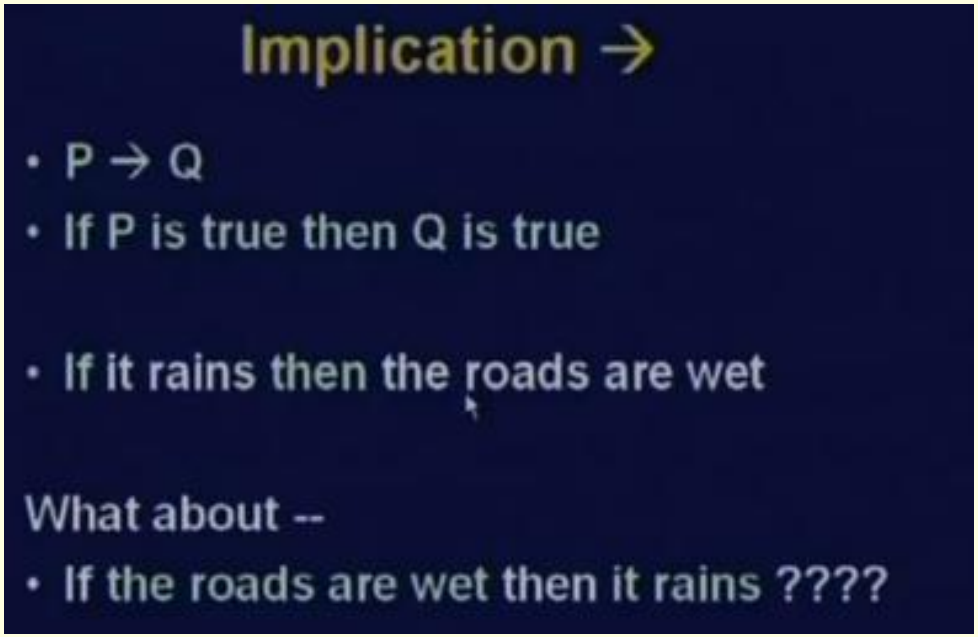
p	q	

p	q	$p \Rightarrow q$
T	T	T
T	F	F
F	T	T
F	F	T

Natural Language Meaning

p	I study hard.
q	I get rich.
$p \Rightarrow q$	If I study hard then I get rich.
	Whenever I study hard, I get rich.
	That I study hard implies I get rich.
	I get rich, if I study hard.

More About Implication

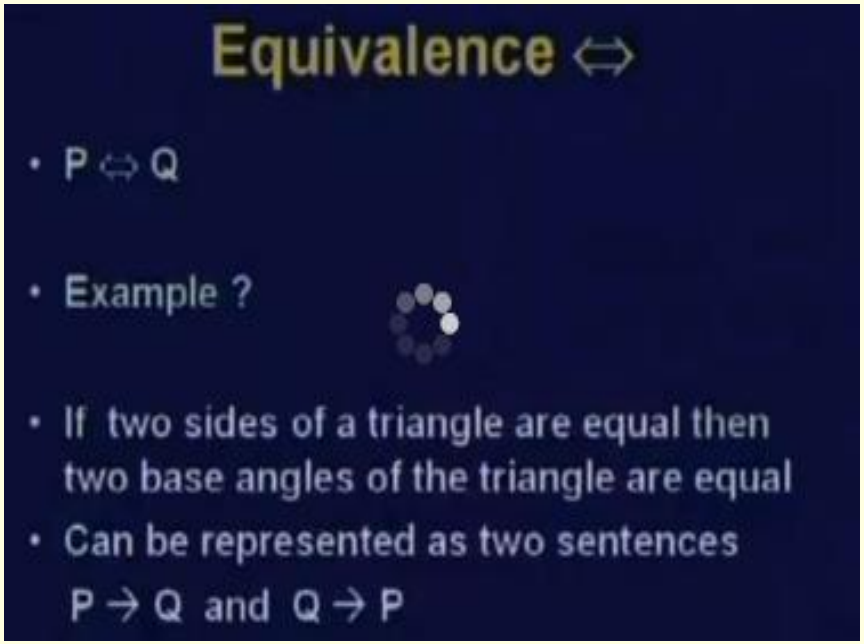


Biconditional $\Leftrightarrow$

- Also known as iff or the biconditional.
- The biconditional, written as $p \Leftrightarrow$, of two propositions is true when both p and q are true or when both p and q are false, and false otherwise.

p	q	$p \Leftrightarrow q$
T	T	T
T	F	F
F	T	F
F	F	T

Biconditional



WFF

The set of sentences or *well-formed propositional formulae* (WFF) is defined as:

- Any propositional symbol is in WFF.
- The nullary connectives, **true** and **false** are in WFF.
- If A and B are in WFF then so is ~A, A v B, A ^ B, A → B and A ↔ B.
- If A is in WFF then so is (A).

Example

The following are well formed formulae

$$p \qquad (p) \qquad p \vee q \qquad (((p \Rightarrow q) \wedge \neg q) \Rightarrow \neg p)$$

Recap: Truth Tables

P	Q	$\neg P$	$P \wedge Q$	$P \vee Q$	$P \Rightarrow Q$	$P \Leftrightarrow Q$
false	*false*	*true*	*false*	*false*	*true*	*true*
false	*true*	*true*	*false*	*true*	*true*	*false*
true	*false*	*false*	*false*	*true*	*false*	*false*
true	*true*	*false*	*true*	*true*	*true*	*true*

Truth Tables for Compound Propositions

- Truth tables may be used to show **interpretations** of compound propositions.
- To draw up a truth table, construct a column for each proposition involved.
- You need 2^n rows for n propositions for all possible ways of setting the propositions to T and F.
- If we have 3 propositions, p; q; r, i.e. we need $2^3 = 8$ rows.
- Next, construct a column for each connective, the most deeply nested first.
- Evaluate each column using values for propositions or previous columns.

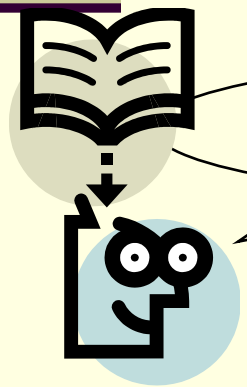
Exercise

p.. (my breakfast is) eggs.

q.. (my breakfast is) cereal.

r.. (my breakfast is) toast.

- The statement ‘my breakfast is either eggs or cereal, and toast’ may be written in symbolic form as ???



WHAT DOES THE TABLE TELL U?

p	q	r	$p \vee q$	$(p \vee q) \wedge r$
T	T	T		
T	T	F		
T	F	T		
T	F	F		
F	T	T		
F	T	F		
F	F	T		
F	F	F		

... So

The truth table gives results for all possible interpretations of p, q and r. The compound proposition is true if

- I eat cereal, eggs and toast; or
- I eat cereal and toast; or
- I eat eggs and toast.

Interpretation - A line in the truth table.

Try this..

- Complete the truth table for $\neg((p \Rightarrow q) \vee (q \Rightarrow p))$

p	q	$(p \Rightarrow q) \vee (q \Rightarrow p)$	$\neg((p \Rightarrow q) \vee (q \Rightarrow p))$
T	T		
T	F		
F	T		
F	F		

- Show that the formula $(p \wedge q) \Rightarrow r$ is satisfiable
- Find an interpretation which satisfies

$$(p \wedge q) \vee (\neg r \wedge s).$$

- Show $p \vee \neg p$ is valid.

Complete the previous slide by referring this site :

http://turner.faculty.swau.edu/mathematics/materialslibrary/truth/

Next..

- Predicate logic